MedAssist AI: Medical Symptom Analysis & Disease Prediction System

Individual Contribution Report (Machine Learning Module)

Field	Details
Name	Kalashree B
Role	Machine Learning Developer
Evaluation Phase	Milestone - 1
Status	Machine Learning Model Completed
Model	XGBoost Multiclass Classifier
Current Accuracy	90.34%

1. Project Overview

MedAssist AI is an intelligent medical symptom analysis and disease prediction system that assists users by predicting possible diseases based on the symptoms selected. The machine learning module was designed to analyze symptom patterns and classify them into multiple disease categories. My responsibility was to implement the end-to-end machine learning workflow, including preprocessing, model development, evaluation, and preparation for deployment.

Machine Learning Workflow

Dataset -> Preprocessing -> Label Encoding -> Train/Test Split

-> XGBoost Training -> Prediction -> Evaluation -> Deployment

2. Individual Contribution

• Studied and analyzed the symptom dataset.
• Verified data quality and handled missing values.
• Encoded disease labels using LabelEncoder.
• Performed stratified train-test split.
• Implemented XGBoost multiclass classification model.
• Generated predictions and evaluated performance.
• Saved the trained model and label encoder for future deployment.

3. Dataset Description

The dataset contains medical symptoms represented as binary features. Each row corresponds to a patient record and each column represents the presence or absence of a symptom. After preprocessing, the dataset consisted of more than fifty thousand records, 377 symptom features and 218 disease classes used for multiclass classification.

4. Data Preprocessing

The preprocessing stage included loading the dataset, identifying missing values, replacing missing values with appropriate defaults, verifying feature consistency, separating input features and target labels, and converting disease names into numeric labels using LabelEncoder. Stratified train-test splitting was performed to maintain the class distribution in both training and testing datasets.

5. Machine Learning Model

XGBoost was selected because it is an ensemble boosting algorithm capable of handling high-dimensional structured data efficiently. The model was configured for multiclass classification using the soft probability objective. Hyperparameters such as learning rate, maximum depth, number of estimators, and evaluation metric were configured to balance prediction accuracy and computational efficiency.

6. Model Evaluation

The trained model was evaluated using standard classification metrics.

Performance Metrics
Accuracy : 90.34%
Precision : 90.16%
Recall : 90.34%
F1-Score : 90.11%

The achieved performance demonstrates that the model is capable of predicting diseases with good reliability while maintaining balanced precision and recall.

7. Technologies Used

Python, Pandas, NumPy, Scikit-learn, XGBoost, Joblib, Google Colab, Jupyter Notebook.

8. Challenges Faced

The major challenges included handling missing values, dealing with multiclass disease prediction involving more than 200 disease categories, optimizing XGBoost training on limited CPU resources, and ensuring proper evaluation across multiple classes.

9. Learning Outcomes

Through this contribution, practical knowledge was gained in data preprocessing, feature engineering, multiclass classification, XGBoost implementation, model evaluation, performance optimization, and machine learning model deployment preparation.

10. Future Work

The next phase of the project includes integrating the trained model with the Flask backend, connecting the frontend symptom selection interface, displaying prediction results with confidence scores, testing the complete system, and deploying the application.

11. Conclusion

The machine learning component of MedAssist AI has been successfully implemented and evaluated. The developed XGBoost model achieved 90.34% accuracy and provides a strong foundation for the intelligent disease prediction system. The trained model is now ready for backend integration and end-to-end application deployment.